SOCIAL EAGLE
learn. build. operate

WEEK 1 · DAY 2

Build a Student Grade Manager

Assignment | learn. build. operate

1. Problem Statement

Your first real program. Build a **Python** program that stores student marks, works out each student's **grade**, and prints the results clearly. It runs in the terminal, no web framework, nothing to install. The interesting part is not the arithmetic; it is how you store the data and what you do with silly input.

Tools: Python only

2. Grading scale to implement

Mark range	Grade
90 to 100	A
80 to 89	B
70 to 79	C
60 to 69	D
Below 60	E

Boundaries are inclusive: exactly 90 is an A, exactly 80 is a B.

3. Requirements

- Keep all your code in a single file named `grade_manager.py`.
- Let the user add as many students as they want, and store them in a data structure you chose on purpose.
- Print a **neatly aligned table**: name, mark, grade.
- Show the **class average**, the highest and the lowest mark.
- Offer a menu loop (add, show, quit) and never crash on bad input.

> **Think it through**
>
> What happens when someone enters 105, or -5, or `abc`? Decide before you write the menu. And pick your storage on purpose: what should your program do about two students both called Priya?

4. Steps to complete

These are hints, not the full answer. Write the pieces yourself.

Step 1: Build it in small pieces

Make a folder, open it in VS Code, and write one function at a time, testing each before moving on. Four small functions beat one long block you cannot debug.

```
students = []              # or whatever you decided on

def get_grade(mark): ...   # mark -> "A".."E"
def add_student():   ...   # ask, validate, store
def show_results():  ...   # aligned table + average, high, low
while True:          ...   # the menu
```

Step 2: Run it and try to break it

```
python grade_manager.py
```

Enter 105, then a word, then nothing at all, then ask for results before adding anyone. Each should be handled, not crash.

5. Final output

```
Name        Mark   Grade
Priya         92       A
Arun          78       C
Divya         59       E
Average: 76.3    Highest: 92    Lowest: 59
```

6. What to submit

- Your `grade_manager.py` on **GitHub**, with the link shared in **WhatsApp**.
- A screenshot of one run with at least three students and the results table showing.
- Two or three lines on how you handled invalid input, posted in the **community**.